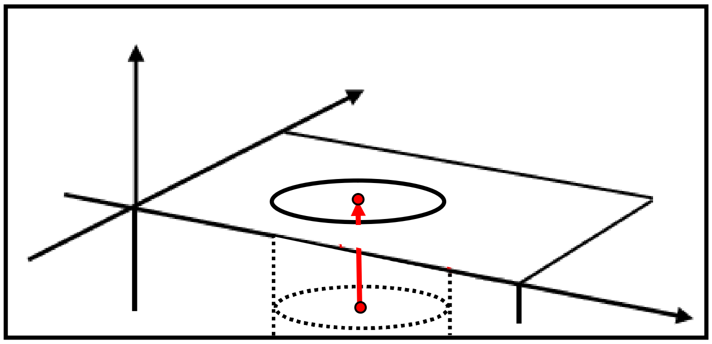
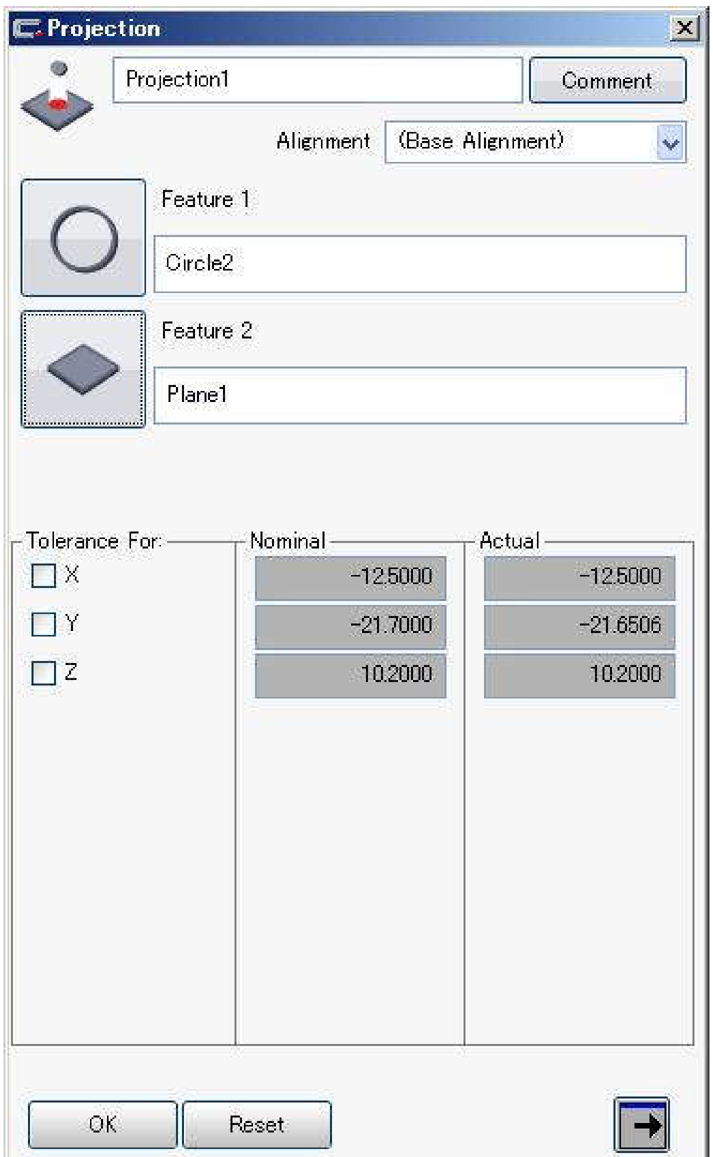
(4) **Projection** ... The feature measured is projected on a plane.
Feature1: A feature to project
Feature2: The projection destination of the Feature1

"Costruction" ⇒ "Projection"

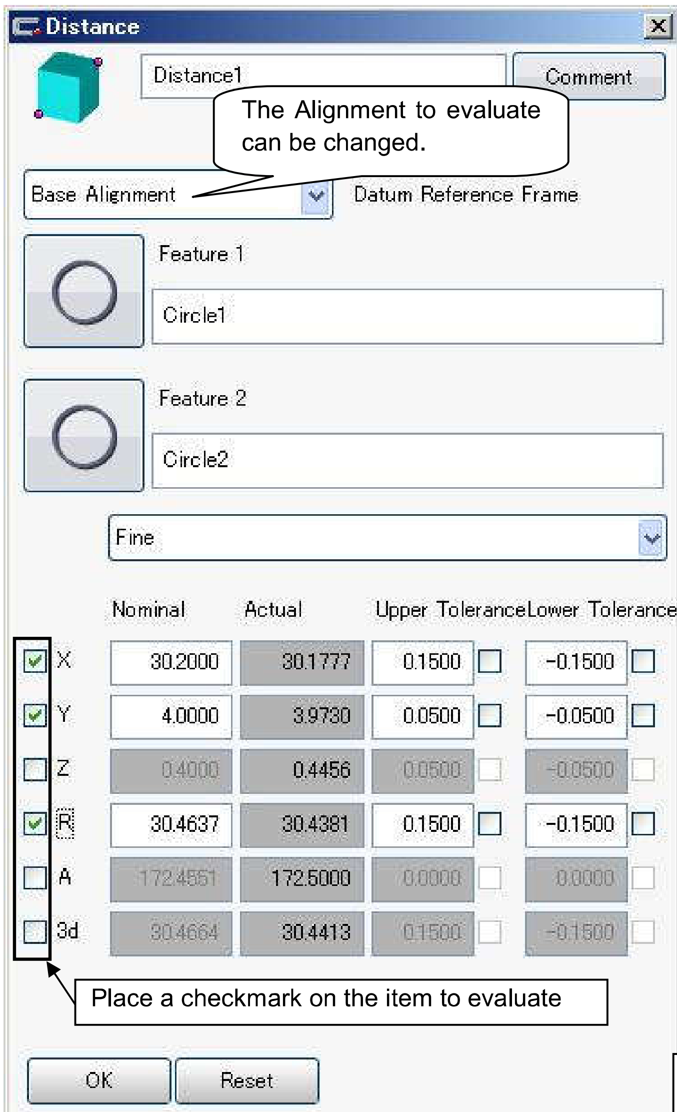


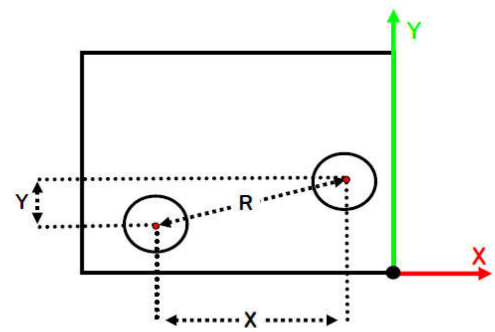


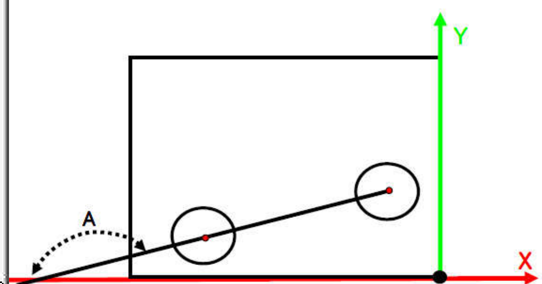
7-2 Evaluation (Distance)

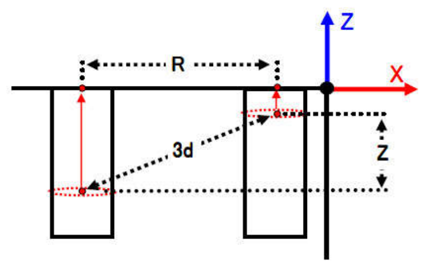
(1) **Distance**...The distance along the alignment axis orientations is collectively calculated.

"Size"⇒ "Distance" ⇒ "Simple distance"









XYZ:
The distance of each axis.

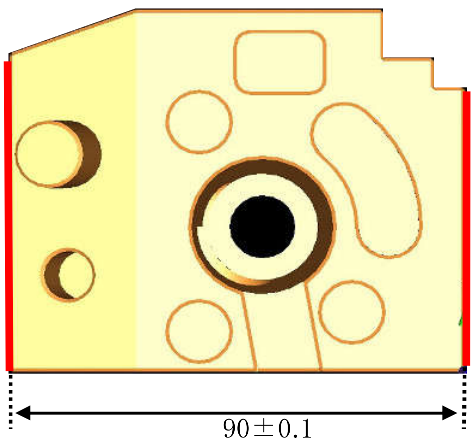
R:
The direct distance of a straight line projected on a measurement plane.

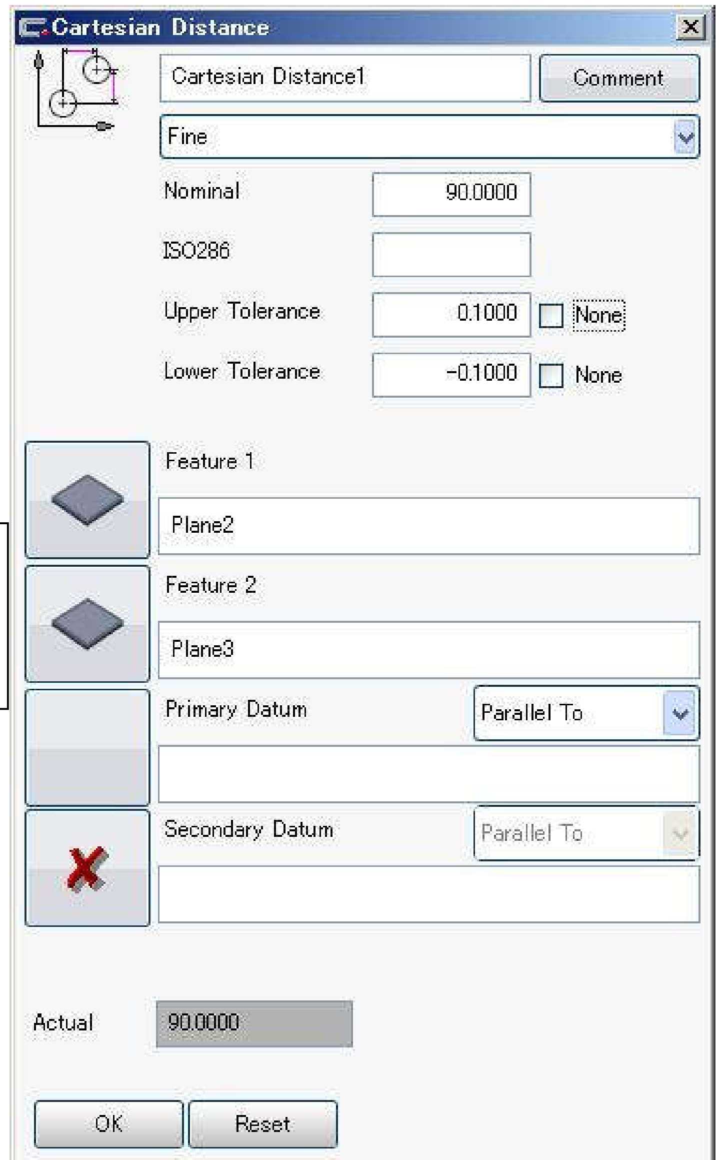
A:
The angle of the line connecting the X-axis of the alignment and two features.

3d:
Three-dimensional direct distance.

(2) Cartesian Distance... Calculation shall be made on the distance between planes such ad grove width, workpiece thickness and so on.

"Size" ⇒ "Distance" ⇒ "Cartesian Distance"





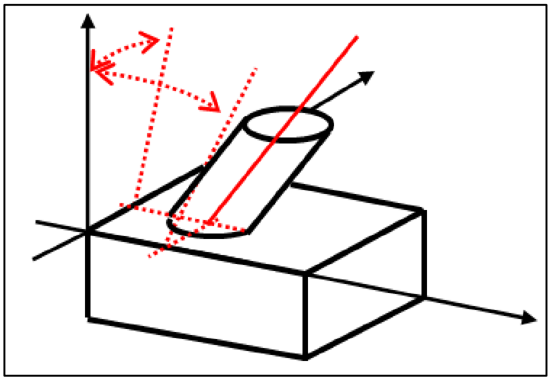
1. Measure plane of two edge faces.
2. Allocate the measured plane to the first and second element.

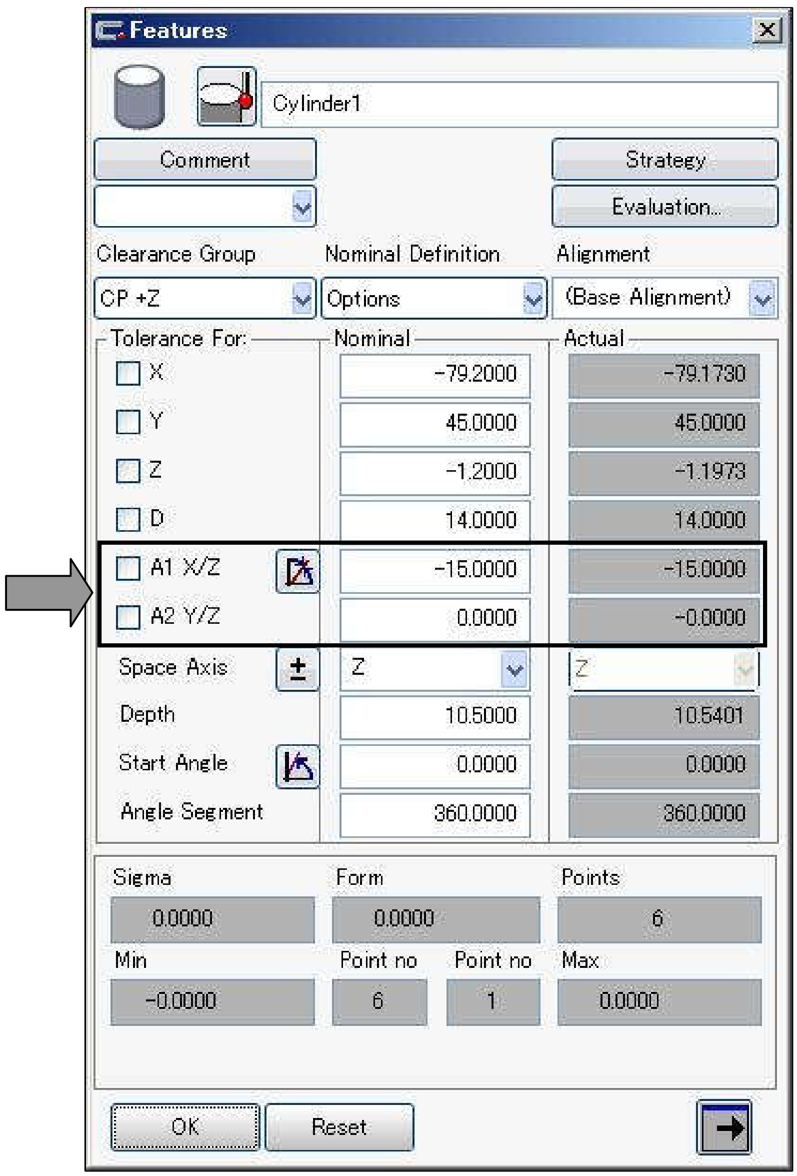
(You can leave the datum column blank.)

Reference!: Vector direction of the plane allocated to the second element shall be the reference and also the distance to the first element.

7-3 Evaluation (Angle)

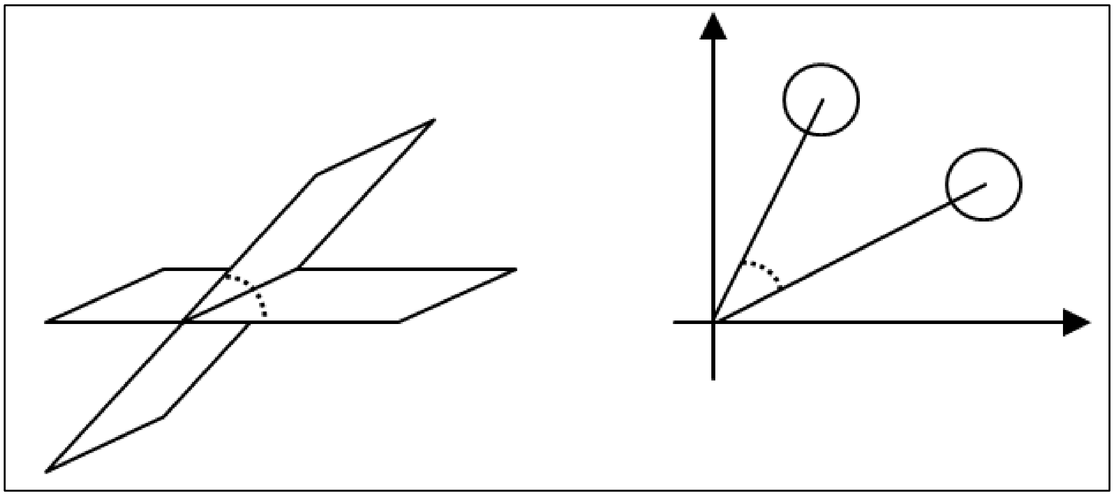
(1) **Projection angle** ... The inclined angle of features seen from a specific direction.
The projection angle is displayed in the feature template.

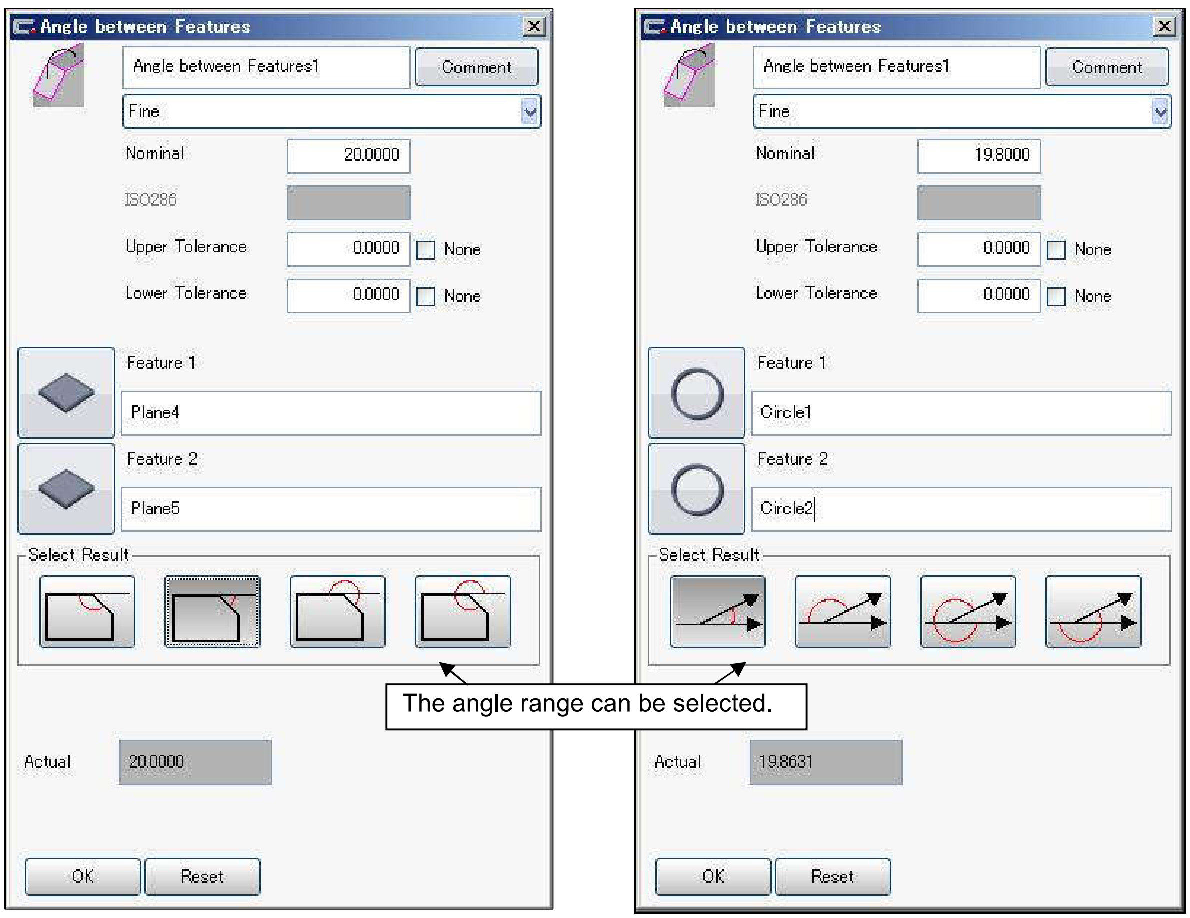




(2) **Angle between features** ... The crossing angle of two features is obtained.
Depending on the combination, it becomes the angle of the two features centering on the home position.

"Form and Location" ⇒ "Angle"





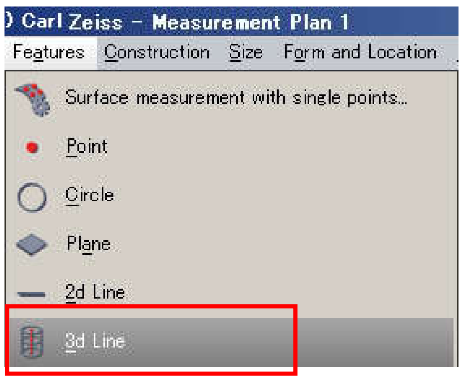
7-4 Recall of Features

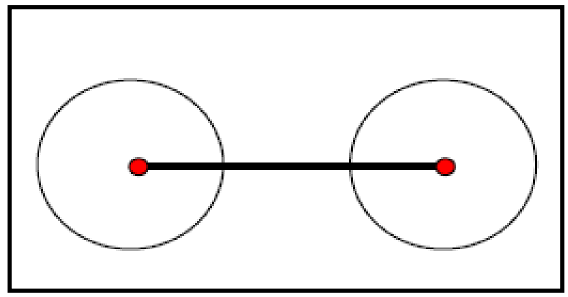
(1) Recall

A coordinate value of a specified feature is recalled.
With this function, a straight line passing through the centers of two circles can be drawn.

1. Left click the 3D line from the [Element] menu.

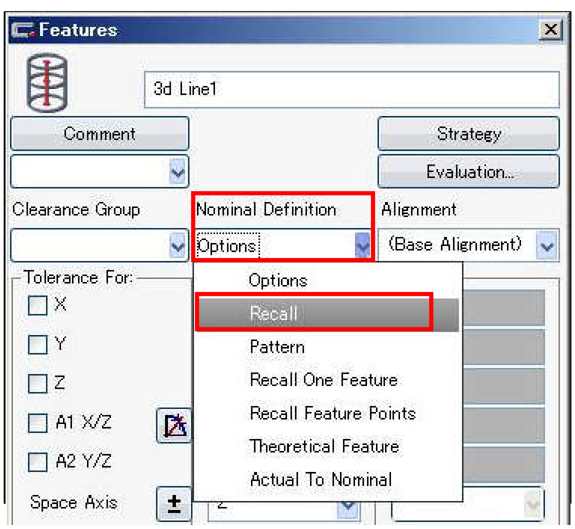
* Be sure to use the 3D line to make a line through calculation as we do this time.

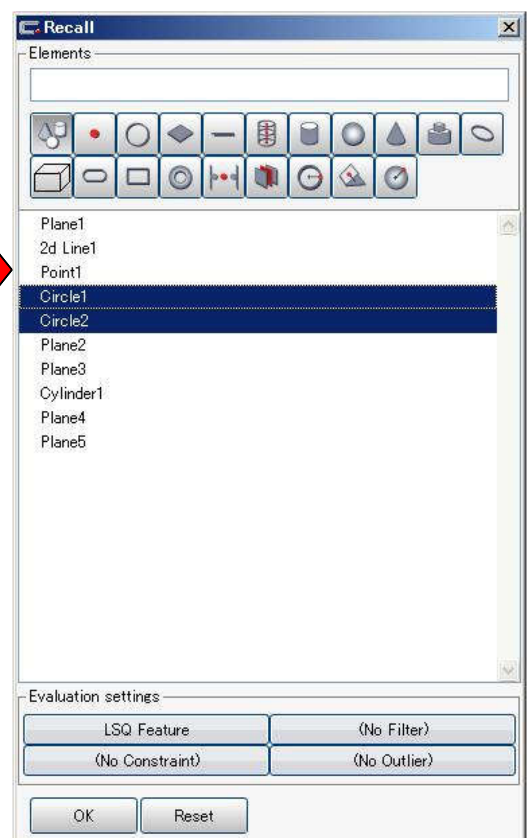




2. Open the added 3D line and left click [Recall] from the menu beneath the [Nominal Definition].

3. Recall window appears and measured elements are displayed.
   From those elements, select two circles to make a line by left clicking the mouse.
   Close the window by OK button.

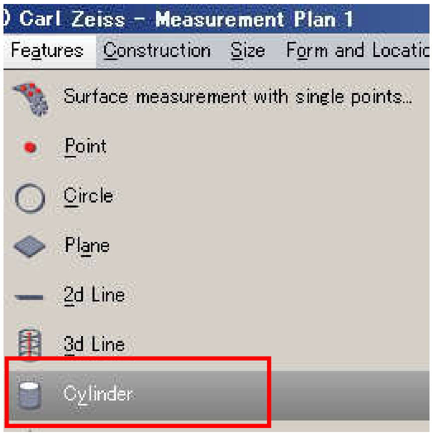


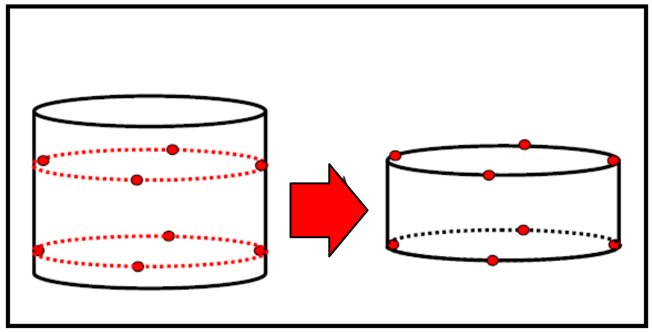


(2) Recall feature points

The point of measurement of a specified feature is recalled.
With this function, a cylinder can be created from two circles.

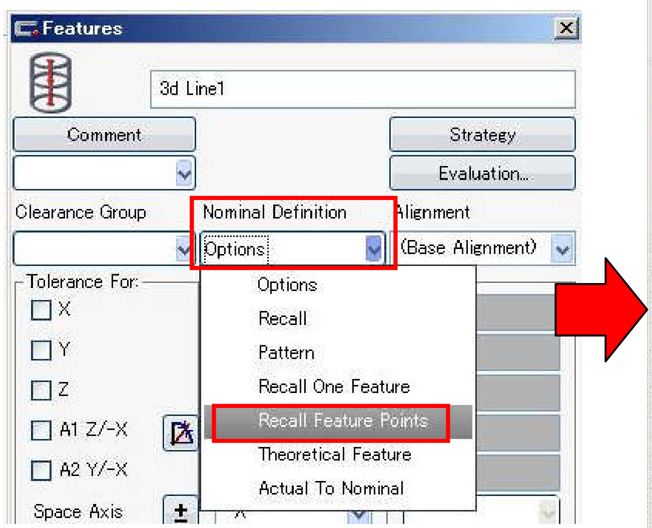
1. Left click the cylinder from the [Feature] of the menu bar.

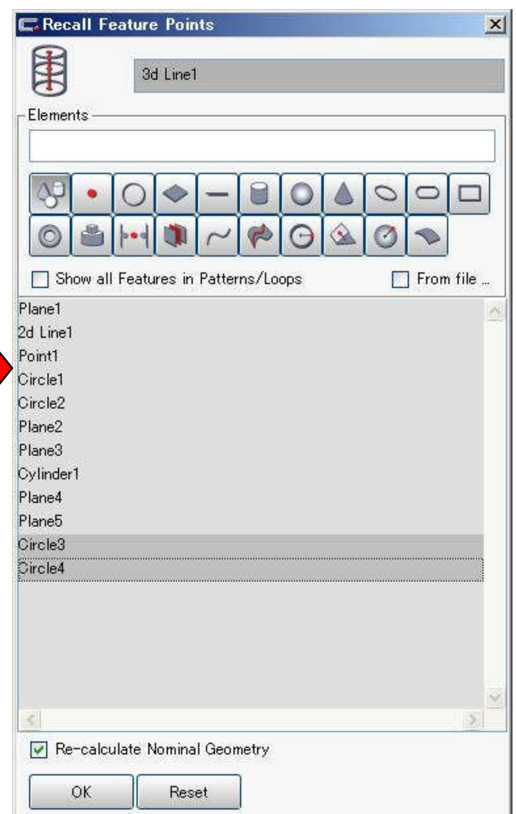




2. Open the added cylinder and left click [Recall Feature Points] from the menu beneath the [Nominal Definition].

3. [Recall Feature Points] window appears and measured elements are displayed.
From those elements, select two circles to make a cylinder by left clicking the mouse.
Close the window by OK button.



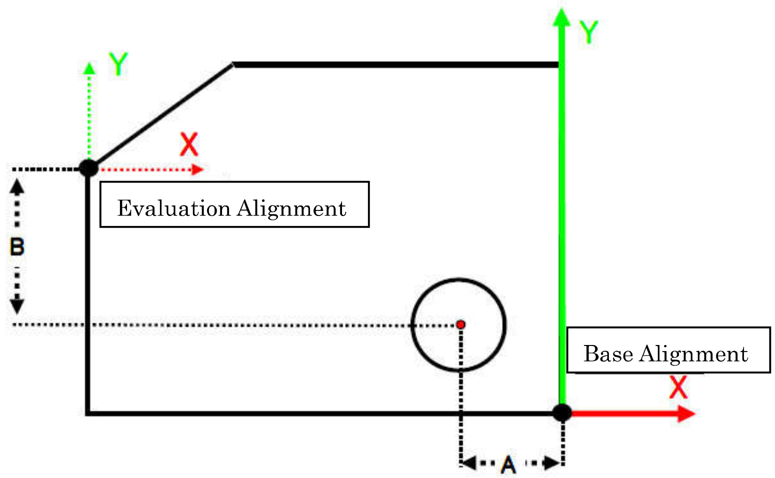


③Recall one element

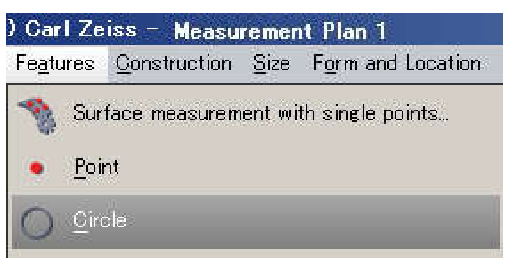
Let us make an element (clone) that has a measured value completely same as that of designated element.

It shall be used when you wanted to evaluate a measured element in a different coordinate system (Evaluation coordinate system)

e.g.) You want to obtain the coordinate value of the hole A. (X from the Evaluation coordinate system) and B. (Y from the Evaluation coordinate system).



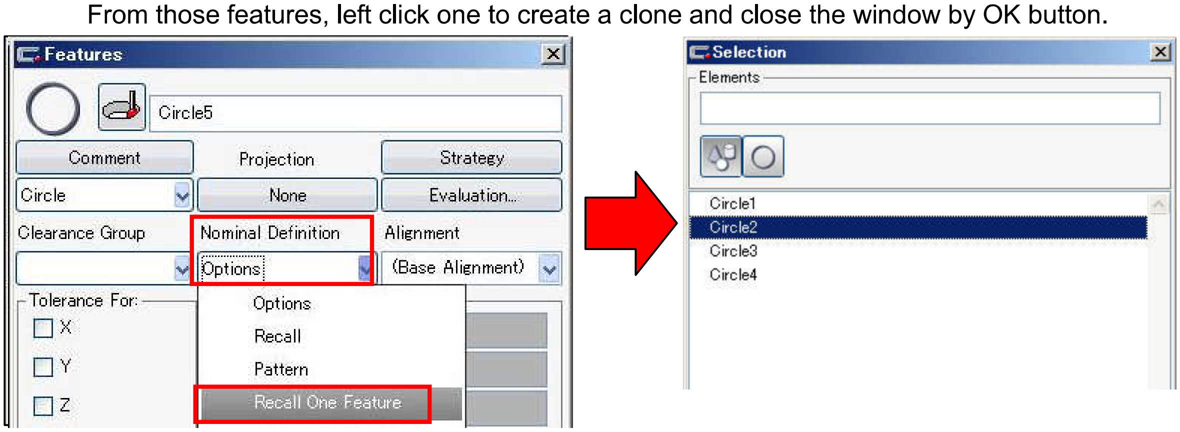
1. From the [Feature] menu, add an feature you want to create a clone.



2. Open the added feature and left click [Recall one feature] from the menu beneath the [Nominal Definition].

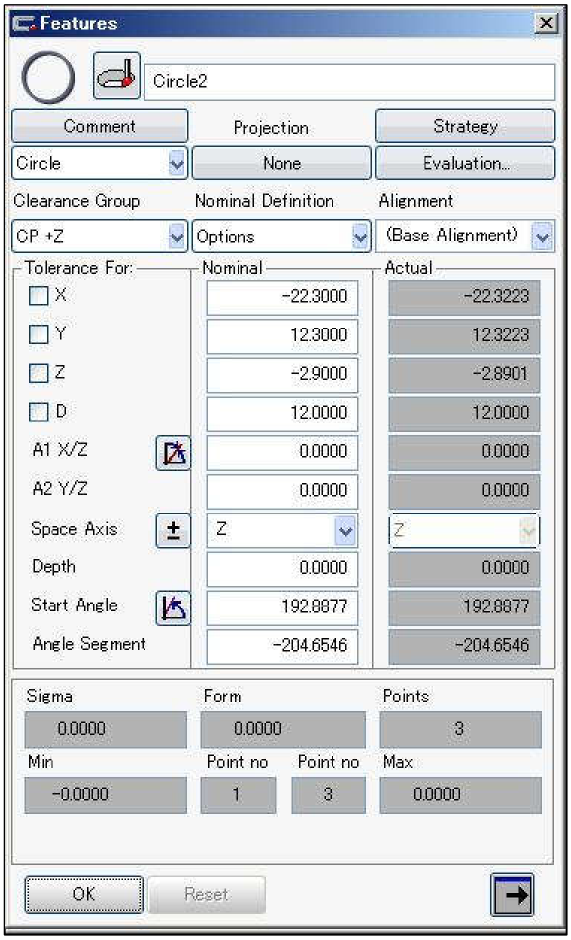
3. Selection menu window appears and measured elements of a type same as the currently opened feature shall be displayed.

   From those features, left click one to create a clone and close the window by OK button.

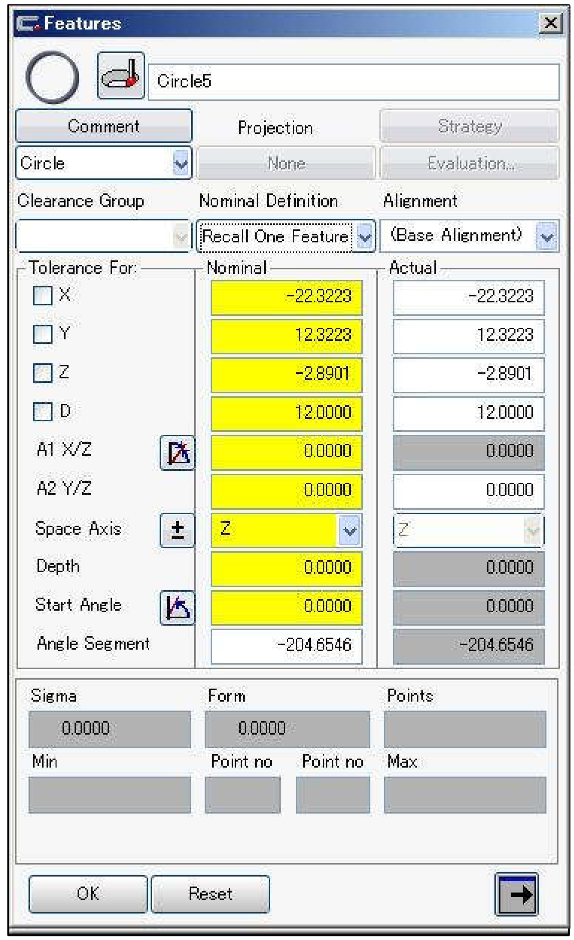


4. A feature having a measured value completely same as that of designated element shall be created (clone)

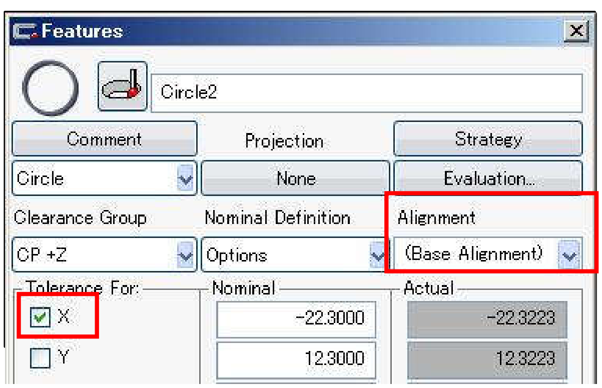
Original feature

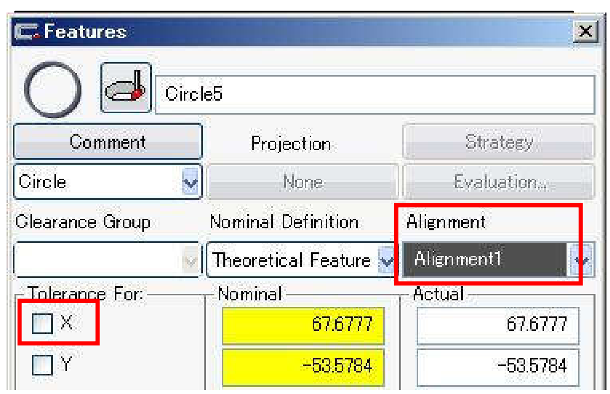


An element created by the recall of one element



5. Let's set a task on changing the coordinate system.

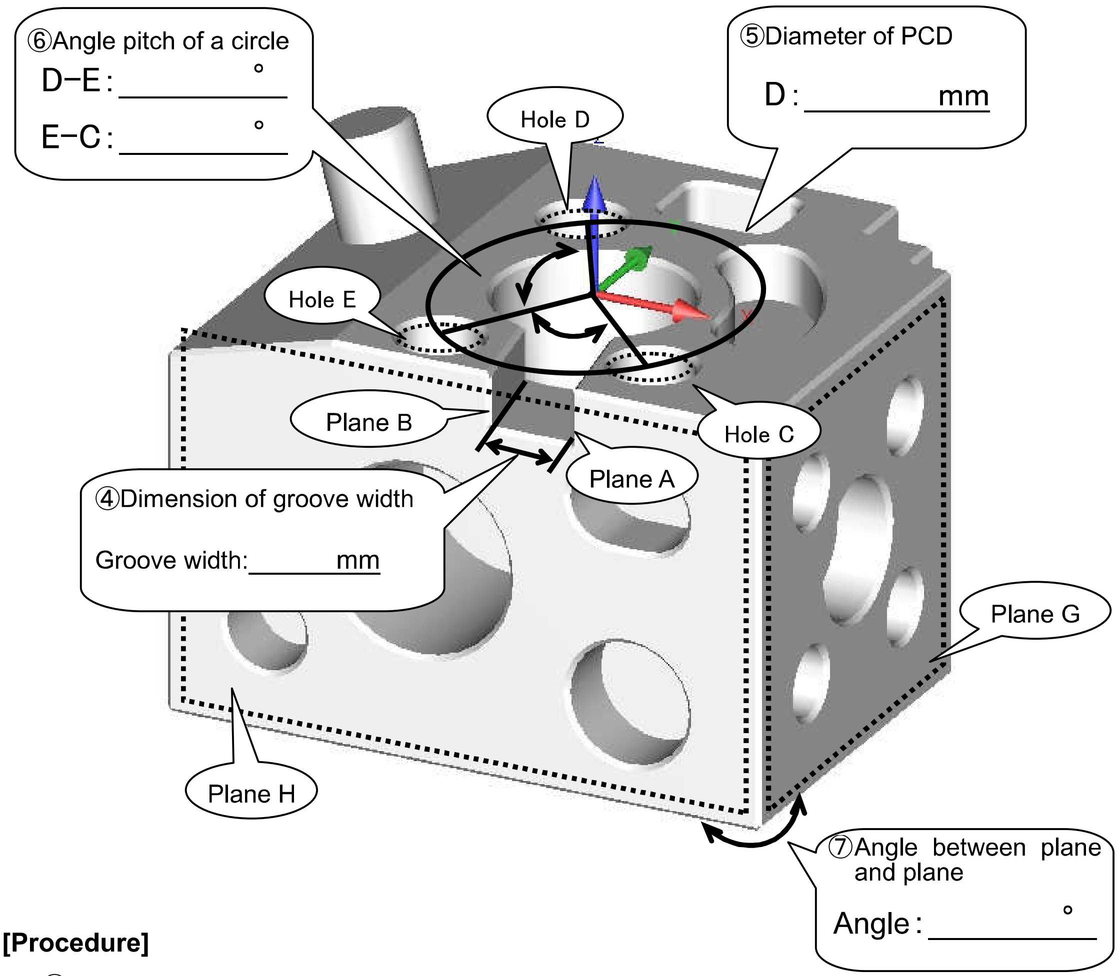




<< Memo >>

Exercise

☆Let's obtain the dimension and angle!☆



[Procedure]

① Open anew a measurement plan.

② Calibrate the Stylus

③ Create the Base coordinate system.

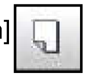
④ Implement plane measurement of opposite planes of the groove to obtain the groove width.
Calculate the groove width from the [cross coordinate].

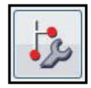
⑤ Measure three holes to obtain PCD.
Create a circle to obtain PCD through [Recall] and evaluate the diameter.

⑥ Obtain an angle pitch of the adjacent hole.
You can obtain the angle pitch by using the angle between elements.

⑦ Measure respective plane to obtain the angle between planes.
Obtain it selecting the icon of the angle you want to obtain.

[Reference values] ④ : 12.0mm, ⑤ : φ50.0mm, ⑥ : DE 120°,EC 75°, ⑦ : GH 90°

(1) Left click the icon [Create New Measurement Plan] 

(2) Implement the stylus calibration 

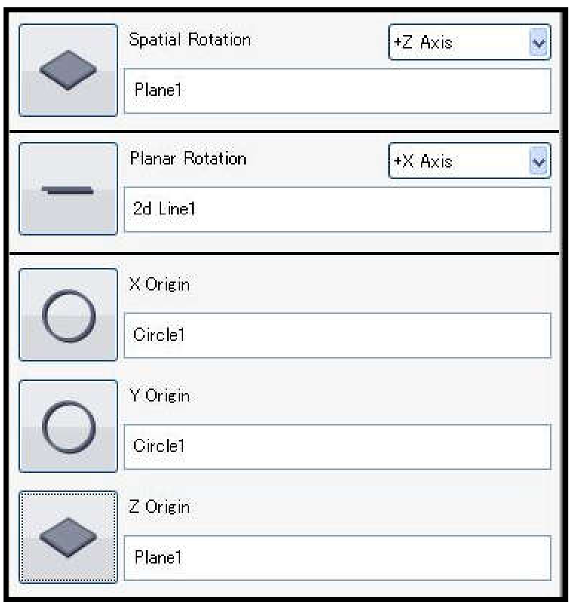
Calibrate the stylus required for measurement.

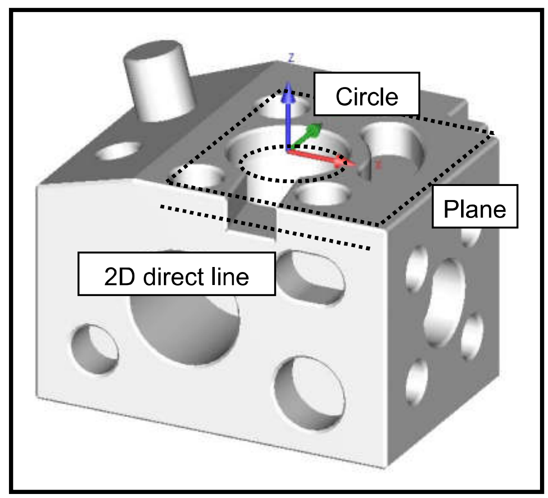
(3) Set the base alignment 

Select [Base / Alignment] – [Create a new base alignment] and left click OK button.
Implement measurement by the following order to set the coordinate system.

- Plane measurement of +Z plane: Space correction, Z-axis origin
- Circle measurement of φ28hole: X, Y axes origin

Line measurement of –Y plane: Rotation correction





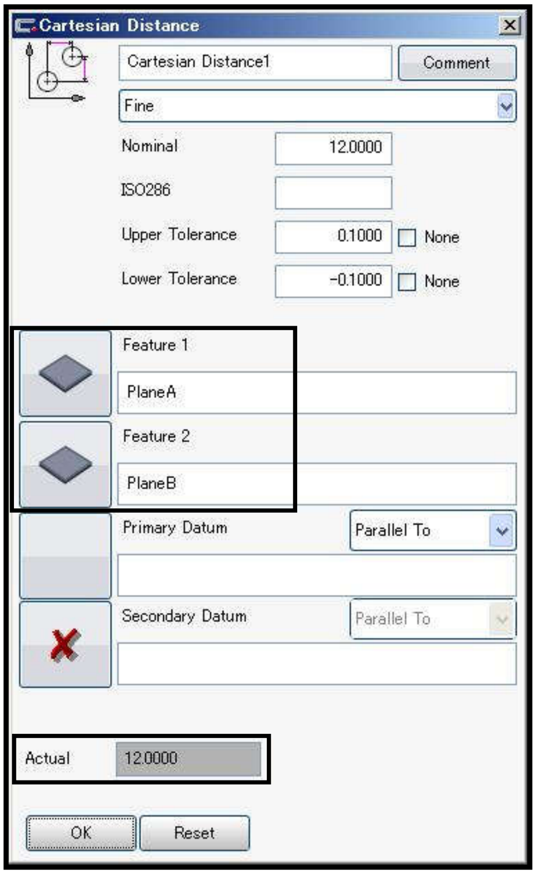
④Implement plane measurement of opposite planes of the groove to obtain the groove width.

1. Implement plane measurement of the plane A and B.
2. Select [Evaluate] – [Distance] – [Cartesian Distance] to add the icon to the Task list.
3. Open the Feature window and select

   Feature 1: Plane A

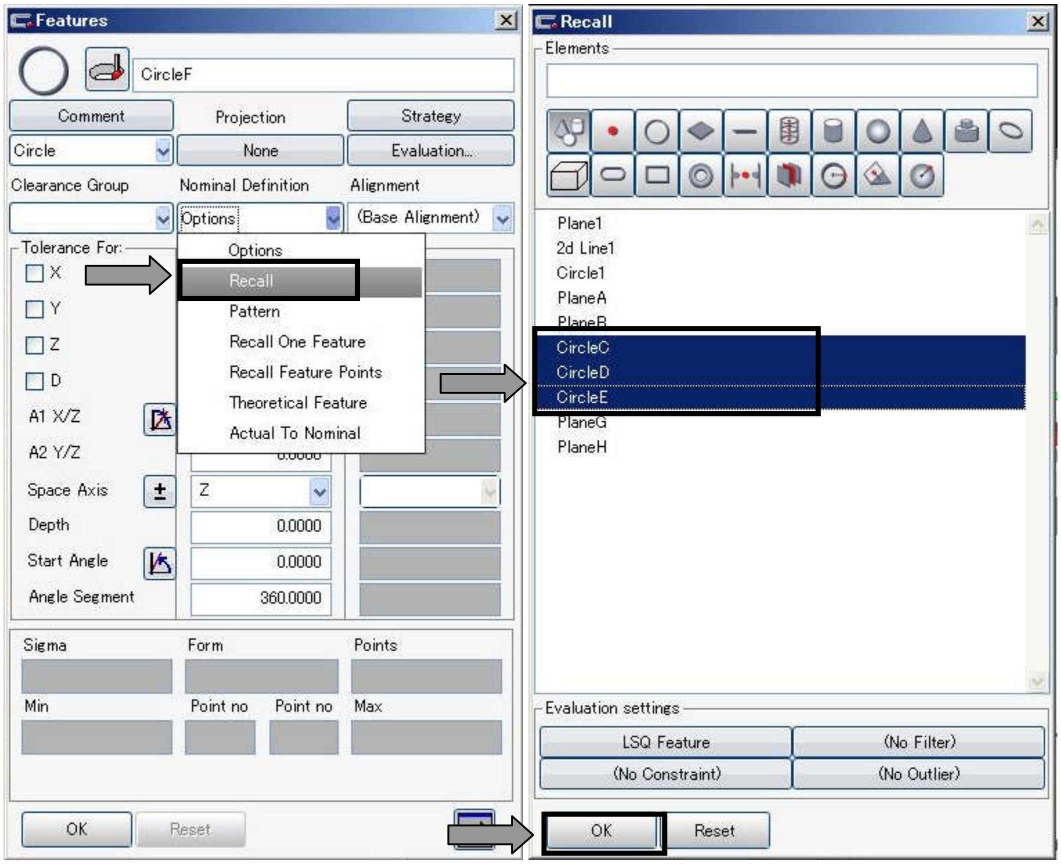
   Feature 2: Plane B

   to obtain the groove width.

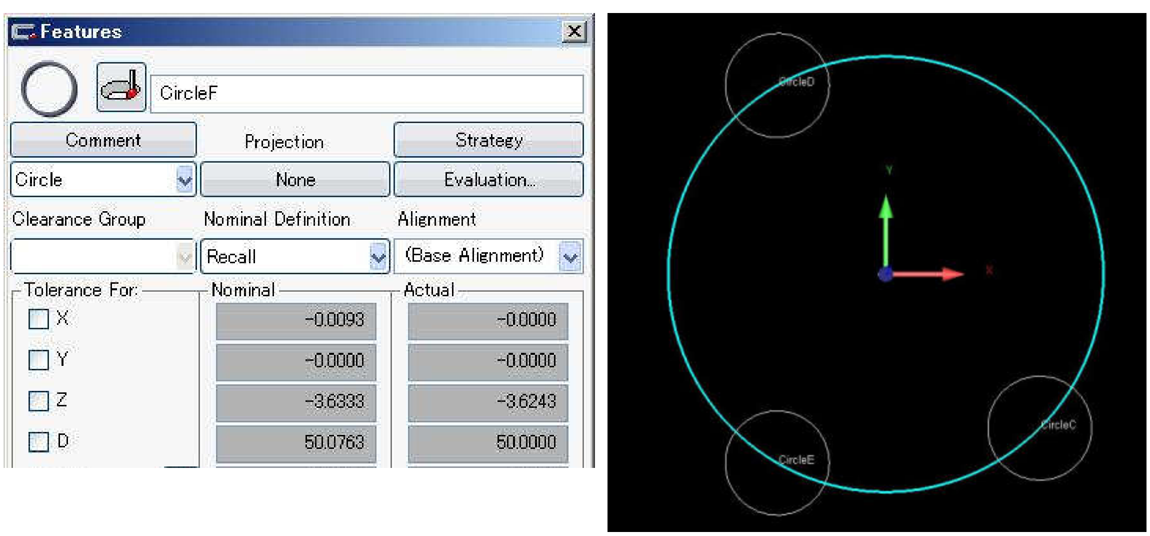


⑤Measure three holes to obtain PCD.

1. Implement circle measurement of the hole C, D and E.
2. Select Feature – Circle to add the icon to the Feature list.
   Open the Feature window and select Nominal Definition – Recall – Plural selection of the circle C, D, E – Close by OK button.



4. A pitch circle is created and the diameter is obtained.



⑥Obtain an angle pitch of the adjacent hole.

1. Select [GD&T] – [Angle] to add the icon to the Task list.
2. Open the Feature window and select

   Feature 1: Circle D

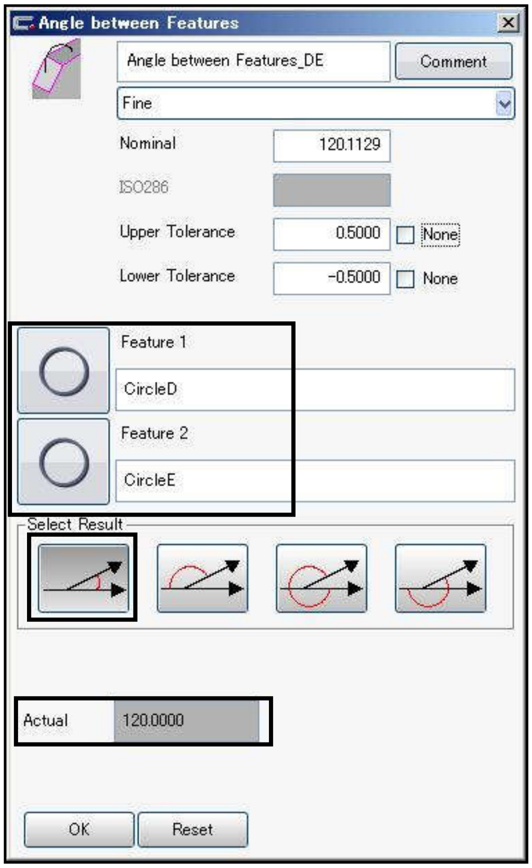
   Feature 2: Circle E

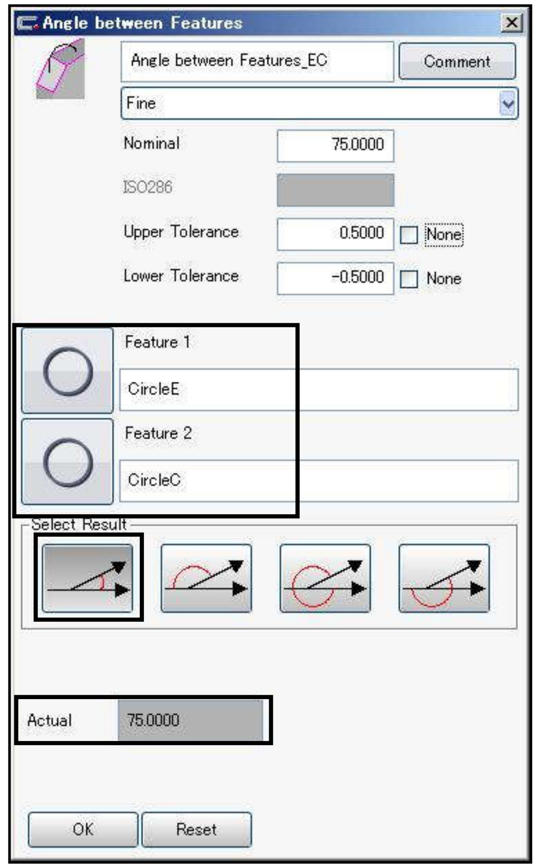
   and select the icon of the angle you want to obtain.
3. In the same manner, select [GD&T] – [Angle], open the Feature window and select

   Feature 1: Circle E

   Feature 2: Circle C

   and select the icon of the angle you want to obtain.

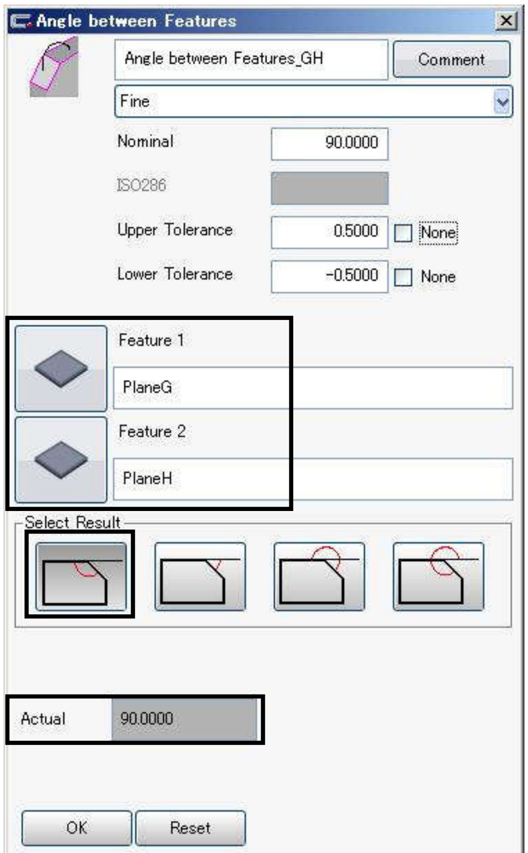




⑦ Implement plane measurement of respective plane to obtain the angle between planes.

1. Implement plane measurement of the plane G and H.
2. Select [GD&T] – [Angle] to add the icon to the Task list.
3. Open the Feature window and select
   Feature 1: Plane G
   Feature 2: Plane H

and select the icon of the angle you want to obtain.



* Please refer to the Appendix 12 to see what angle is obtained.

<< Memo >>

GD&T

Startup and Exit of CMM

Stylus calibration

Measurement of features

Creation of Alignment

STEP5

-Setting Characteristics(Drawing inspection items)-

We intorduce you functions to obtain inspection items of the drawing.

Construction

Distance, Angle, Recall

Geometric tolerance

CNC Measurement

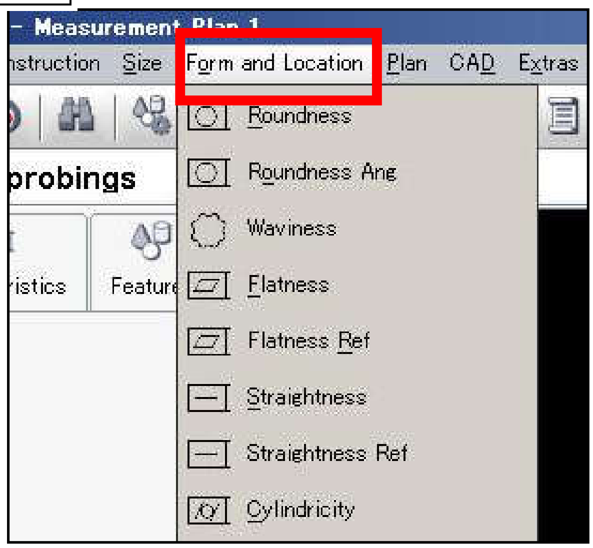
1. Setting the Geometric Tolerance

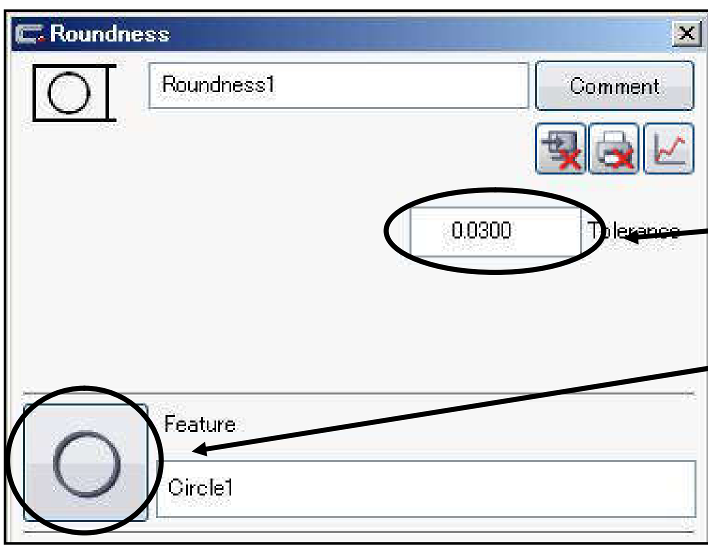
① Profile deviation

Major geometric tolerance examples

- Roundness
- Flatness
- Cylindricity
- Straightness

Let us evaluate whether the profile of an object is same as the image.
You can output the result of those geometric tolerances by just adding from the menu and allocating the element to evaluate.





Type in the tolerance

Allocate the Feature to evaluate

* Evaluation method of the element is the Minimal region method (Chebyshev element).
You can change the method as necessary

Reference: Appendix 13 “Display method of the profile plot”